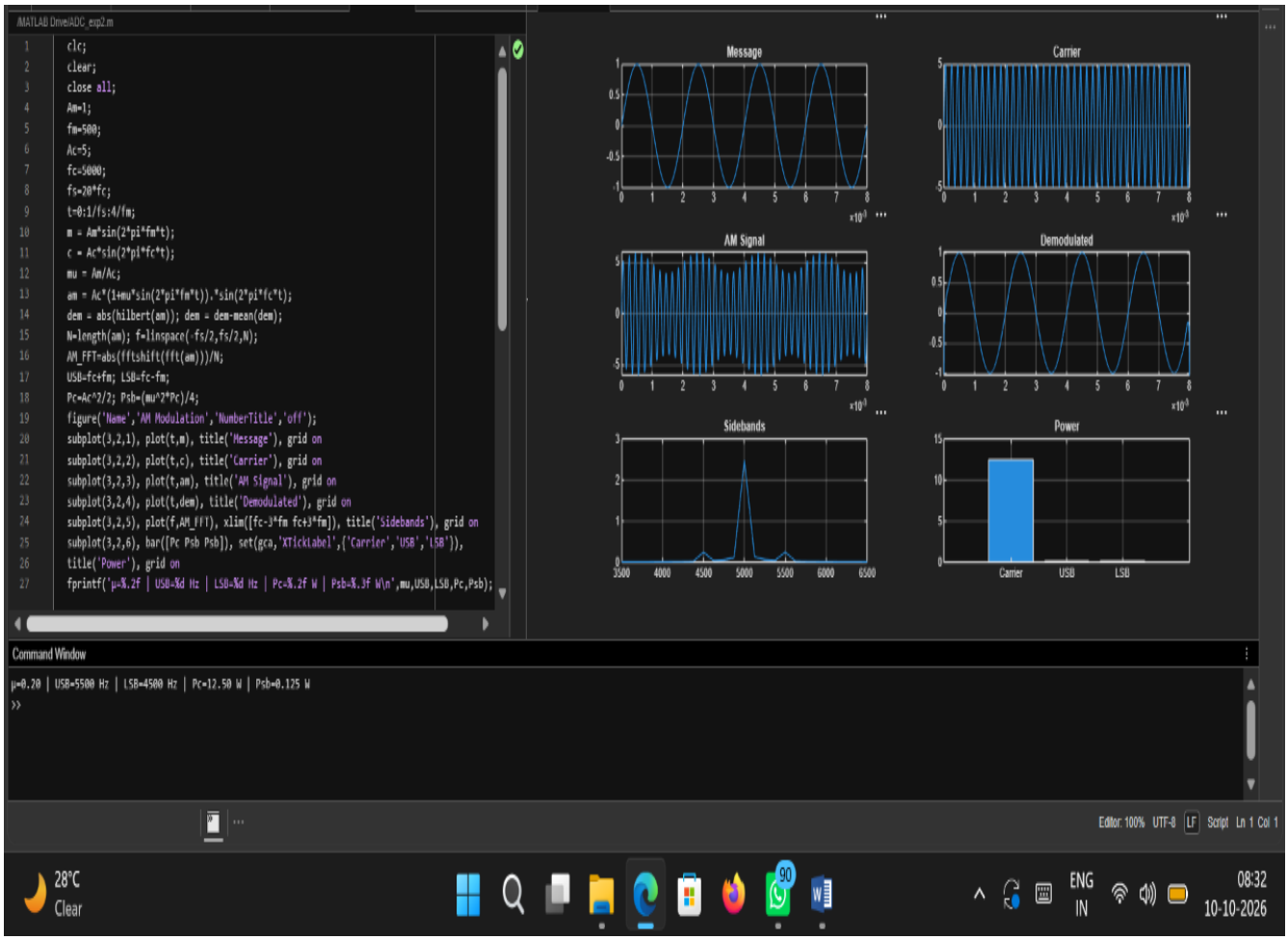
# Ex.No:2: CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

## Objective 1:




# Code:

```
clc;

clear;

close all;

Am=1;

fm=500;

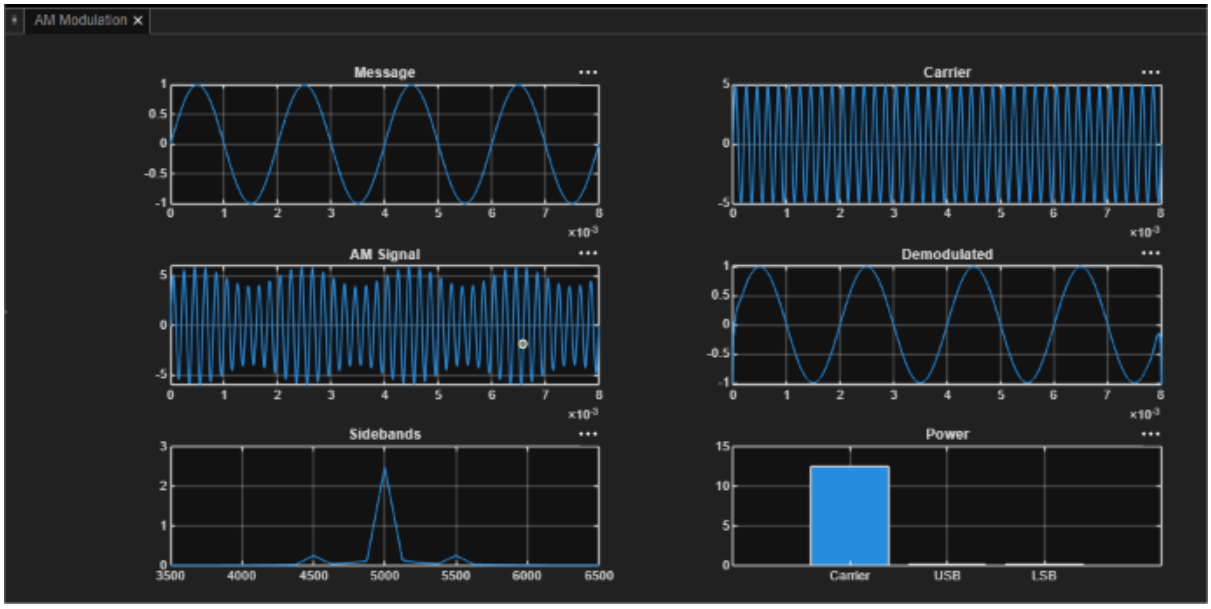
Ac=5;

fc=5000;

fs=20*fc;
```

```
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
USB=fc+fm; LSB=fc-fm;
Pc=Ac^2/2; Psb=(mu^2*Pc)/4;
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1), plot(t,m), title('Message'), grid on
subplot(3,2,2), plot(t,c), title('Carrier'), grid on
subplot(3,2,3), plot(t,am), title('AM Signal'), grid on
subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on
subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'),
grid on
subplot(3,2,6), bar([Pc Psb Psb]),
set(gca,'XTickLabel',{'Carrier','USB','LSB'}),
title('Power'), grid on
fprintf('μ=%.2f | USB=%d Hz | LSB=%d Hz | Pc=%.2f W | Psb=%.3f
W\n',mu,USB,LSB,Pc,Psb);
```

# Output:


AM Modulation
Message
Carrier
AM Signal
Demodulated
Sidebands
Power
Carrier
USB
LSB